

State of the art in eukaryotic gene prediction

Tyler Alioto and Roderic Guigó

1 Introduction

Computational gene prediction is the cornerstone upon which a genome annotation is built, as gene prediction is usually the first step taken toward the annotation of a newly sequenced genome. This is largely due to the fact that computational identification of the entire repertoire of genes in a genome is vastly more economical than the experimental identification of each and every gene, or for that matter, even a single gene. Apart from the economic driving force behind the development of the gene prediction field, there also exists a fundamental scientific or intellectual driving force: in order to precisely delineate the gene structures within anonymous genomic sequences, we must be able to accurately model, and therefore understand, individually and collectively the mechanisms of transcription, splicing, mRNA maturation, nonsense-mediated decay, translation and even non-coding RNA regulatory circuits.

The interplay between prediction and experimentation should be seen as hypothesis driven, not data driven, biological research. Each gene prediction is a hypothesis waiting to be tested and the results of testing then inform our next set of hypotheses. It is really no different than the early days of gene-finding. Ever since genes were defined as the hereditary units that confer traits or phenotypes to organisms, their study has been essential to the study of biology. The discoveries that genes reside in deoxyribonucleic acid (DNA), are transcribed into ribonucleic acid (RNA) and (in many cases) then translated into polypeptides spurred the rapid development of molecular biology, which revolves around trying to understand the function of genes

Tyler Alioto

Center for Genomic Regulation, c/ Dr. Aiguader, 88, E-08003 Barcelona, Spain, e-mail: tyler.alioto@crg.es

Roderic Guigó

Center for Genomic Regulation, c/ Dr. Aiguader, 88, E-08003 Barcelona, Spain, e-mail: roderic.guigo@crg.es

at the molecular level. Thus it has become requisite that their coding sequences and, by necessity their physical locations within the genome and intron-exon structures, be determined.

Methods for finding genes have evolved since the early days of genetics. In the pre-genomic age, genetic maps were constructed by analysis of phenotypic segregation (either natural traits or mutant phenotypes) in large pedigrees or through series of genetic crosses. In the post-genomic age, the gene finding problem has largely turned into a computational one. The task can now be stated as follows: given a DNA sequence, perhaps a chromosome or entire genome, what are the precise boundaries and exonic structures of all of the genes?

In prokaryotes and some simple eukaryotes, the computational solution is a relatively simple task: to identify long open reading frames (ORFs) that, due to their length, are likely to code for proteins. The precise start codon can often be identified using simple rules such as choosing the ATG that maximizes the length of the ORF. The presence of other signals such as a Pribnow box (TATAAT consensus), the -35 sequence or ribosomal binding sites can be used to refine the prediction of the transcriptional and translational start sites. Furthermore, codon bias is often used to deduce the correct frame for overlapping ORFs. The accuracy of prokaryotic gene finders is upwards of 90% for both sensitivity and specificity. GLIMMER (Salzberg *et al.*, 1998) is perhaps one of the most accurate prokaryotic *ab initio* gene finders. It uses an interpolated Markov model (IMM), discussed later in this chapter. GeneMark (Borodovsky and McIninch, 1993) is another successful prokaryotic (and now also eukaryotic) gene finder which pioneered the use of the 3-periodic Markov model for exon recognition that forms the basis of almost all modern gene predictors.

Eukaryotes, on the other hand, are more complex and pose a much greater challenge. First of all, their genomes can be orders of magnitude larger, and much of their DNA sequence does not code for proteins. For instance, only 3% of the human genome codes for proteins. Second, genes are almost always split into smaller coding sequences (exons) by intervening non-coding sequences (introns) which are spliced out of pre-messenger RNA by a ribonucleoprotein complex called the spliceosome to form a mature mRNA (see Figure 1). Introns can sometimes be very large (> 100kb), making the search for exons like trying to find a needle in a haystack. Not to mention the fact that due to alternative splicing, multiple mature transcripts can be derived from one pre-mRNA. Alternative transcription start sites are also quite common. Genes can also be interleaved, overlapping, or nested, adding to the complexity.

Thus, for simplicity's sake, gene finding efforts to date have mainly focused on finding the genomic coordinates corresponding to a single protein-coding sequence per non-overlapping genomic locus. UTRs have largely been ignored as well as non-canonical splice sites (including U12 introns). That said, we must take note that this operational definition of a gene may have to be modified as our understanding of the transcriptional activity of the genome increases. A large proportion of the transcriptional activity in eukaryotic genomes, according to the results of new experimental techniques, appears not to code for proteins. These transcripts of unknown function,

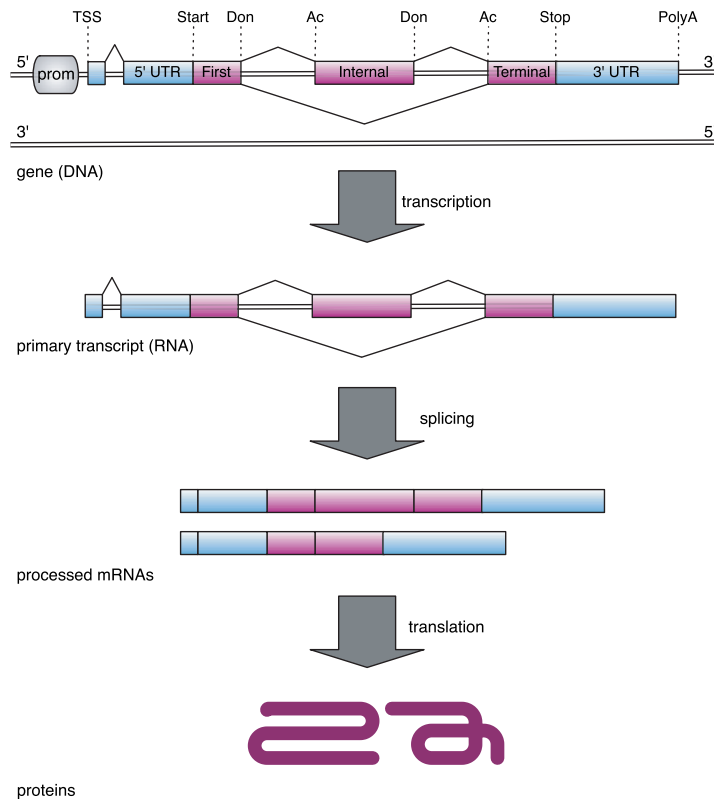


Fig. 1 Typical eukaryotic gene structure. Protein-coding genes are typically interrupted by non-coding sequences called introns, which are spliced out of the primary transcript (sometimes alternatively) to produce one or more mature messenger RNA products, which are then translated starting at the start codon and ending at the first in-frame stop codon

polyadenylated and non-polyadenylated, sense and antisense, overlapping and interleaved with protein coding genes, are distorting what once seemed to be a clear concept of a “gene” (Gingeras, 2007).

For clarity, we will assume the operational definition but, where possible, highlight cases in which some of the complexities of transcription, RNA processing and translation are starting to be addressed. Even with these simplifying assumptions, gene finding programs exhibit far from perfect performance, thus we will refer to computational gene finding as “gene prediction” reflecting the still-necessary step of validating the gene models predicted by these programs.

In the next section we will introduce the basic principles of gene prediction, namely signal and content detection, and in the following section, we will illustrate how they are incorporated into modern eukaryotic gene finders. We will also discuss the development of more sophisticated frameworks for combining signal and con-

tent sensors with diverse sources of information such as phylogenetic conservation and genomic alignments of expressed sequences.

2 Classes of Information

We will begin by introducing the main sources of information that have been traditionally used to find genes. Then in Section 3 we will outline how this information can be captured and incorporated into gene model predictions. Information can be divided logically into two main categories, extrinsic and intrinsic, based on whether or not the information can be derived solely from the target genome sequence.

2.1 Extrinsic Information

Extrinsic information includes any source of evidence that is not itself a genome sequence. In general, we refer to expressed sequence such as cDNAs, expressed sequence tags (ESTs) or the sequence of their protein products as extrinsic information. Gene prediction methods which do not use this information are referred to as *de novo* methods.

Homology information can be used in several ways according to quality and completeness. If the homologous sequence is derived from the same species and locus as the target sequence, then a spliced alignment approach often suffices to accurately map the region of homology. If the homologous sequence is full-length, such as a full-length cDNA sequence, and the boundaries of the transcript coincide with canonical splice sites, the coordinates of the genomic alignment represent the gold standard of gene annotation to which all other methods are compared. Determination of the start and stop codons then usually entails finding the longest open reading frame, although on occasion the true start codon is not the first methionine codon encountered. The presence of a Kozak consensus sequence ([A/G]XXAUGG) (Kozak, 1981) can help distinguish true start codons from other potential start codons nearby.

Although BLAST (Altschul *et al.*, 1990) is often used to roughly locate a gene within a genomic sequence using a homologous sequence, precise mapping of homologous sequences to the genome is ideally performed by programs specifically designed to perform spliced alignments. Procrustes (Gelfand *et al.*, 1996), EST_GENOME (Mott, 1997), sim4 (Florea *et al.*, 1998), BLAT (Kent, 2002), GMAP (Wu and Watanabe, 2005), and Exonerate (Slater and Birney, 2005) are a few such examples. Genewise (Birney and Durbin, 2000; Birney *et al.*, 2004) is another program that aligns proteins to the genome. All such spliced aligners use either a basic model (terminal dinucleotide consensi) or more sophisticated models (such as position weight matrices/arrays) of splice junctions and introns.

If the region of homology is incomplete or of lower quality, then the preferred approach is to extend the spliced alignment with *ab initio* gene prediction. This ap-

proach is generally implemented as a stepwise pipeline such as in ENSEMBL (Hubbard *et al.*, 2002) or UCSC genes (Hsu *et al.*, 2006). However, EST and cDNA alignments may also be incorporated directly into gene predictions through extensions to gene predictors like Twinscan (Wei and Brent, 2006), or, as is becoming more common, by “combiner” programs. At low levels of identity, BLAST high-scoring-pairs (HSPs) can either be used to weight predicted exons in a non-probabilistic way or may be incorporated into gene prediction probabilistically using pair hidden Markov models (see below).

2.2 Intrinsic Information

De novo gene predictors are programs that predict the exon-intron structures of genes using the sequences of one or more genomes as their only input. The term *ab initio* is used strictly for *de novo* gene predictors that do not use informant genomes, and more or less means “from first principles.” The most “*ab initio*” of gene prediction programs would be a program that simulates the transcription, splicing and postprocessing of a transcript using only the information available to the cell. Such a simulator, if successful, would truly demonstrate our understanding of the molecular mechanisms and dynamics of gene expression. However, our understanding at this point is at best rudimentary and we must rely on metrics derived from many examples of genes with known exonic structures. These informative metrics can be categorized as either signal sensors or content sensors.

2.2.1 Signals

The signals in which we are interested are nucleic acid sequence motifs that are recognized by the cellular machinery responsible for transcribing, processing and translating messenger RNA molecules. The minimal set of signals that describes the structure of a coding sequence (CDS) include the start and stop codons and, if there is more than one exon in the coding part of the transcript, the donor and acceptor splice sites for each intron present. The acceptor site may be sometimes be defined as a composite of branch site, poly-pyrimidine tract and the acceptor junction signals. Additional signals that may be taken into consideration are splicing enhancer and silencer elements, transcription start and termination sites, poly-adenylation signals, and even proximal and distal promoter sequences.

Many of these signals can be modeled as simple position weight matrices, or PWMs (alternatively known as position specific scoring matrices or position specific probability matrices). PWMs attempts to capture the intrinsic variability characteristic of sequence patterns and are usually derived from a set of aligned sequences which are functionally related. PWMs simply tabulate the frequency with which each nucleotide is observed at each position. Formally, from a set S of n aligned sequences of length l , $s_1, \dots, s_n$, where $s_k = s_{k1}, \dots, s_{kl}$ (the s_{kj} being one of A, C, G,

T in the case of DNA sequences) a Position Weight Matrix, $M_{4 \times l}$ is derived as

$$M_{ij} = \frac{1}{n} \sum_{k=1}^n I_i(S_{kj})$$

$$i \in [A, C, G, T]$$

$$j = 1 \cdots n$$

$$\text{where } I_i(q) = \begin{cases} 1 & \text{if } i = q, \\ 0 & \text{otherwise.} \end{cases}$$

This matrix is usually converted to a frequency or probability matrix with the sum of each column equal to one. A novel sequence can now be searched for this motif by moving a window the size of the motif across the sequence and for each position of the matrix summing the frequencies corresponding to each nucleotide observed. A score is obtained where the higher the score the better the match. However, scores from different matrices are difficult to compare and selecting a proper threshold becomes rather empirical. The solution to this problem is to use a “background” model. Background frequencies could be equiprobable nucleotide frequencies with 0.25 for each A, C, G and T, or the frequencies may be derived from the true genome-wide nucleotide frequencies or perhaps from the local context of the true sites. The likelihood of a sequence belonging to the category of the motif becomes the product of the probabilities of the observed nucleotides occurring in each position of the motif divided by the product of the probabilities of the background nucleotides in each position of the motif. If we then take the log of this ratio, called the log-likelihood ratio, then sequences with scores above zero can be interpreted as being more likely to be an instance of the motif, while those that score below zero are not likely to be. If we store the log likelihood ratio for each position of the motif in the matrix, then we may simply take the sum of these ratios at each position to be the score of the entire motif. This method is illustrated in Figure 2 using the U12 branch point PWM as an example (U12 introns, which comprise only a fraction of a percent of all human introns, are spliced by the minor U12 snRNP-containing spliceosome).

Dependencies between adjacent positions can be captured in a weight array matrix (WAM) model. The probabilities in the matrix are now calculated as conditional probabilities, where the probability of a sequence $S = s_1 \cdots s_n$ being an instance of a particular motif is

$$P(S) = P(s_1)P(s_2|s_1)P(s_3|s_2)P(s_4|s_3)P(s_5|s_4) \cdots P(s_n|s_{n-1})$$

where $P(s_i|s_j)$ is the probability of nucleotide s_j in position k given that nucleotides s_i is at position $k - 1$. Log-likelihood ratio scores can also be computed, by calculating the probability of the sequence S under some background model.

This type of dependency, where the state at one position is conditioned only on the state immediately preceding it (in space or time) fulfills the Markov assumption. Thus, these models can also be thought of as 0-order and 1st-order Markov Chains, respectively, where the order refers to the number of immediately preceding nucleotides on which the probability of observing a particular base is conditioned.

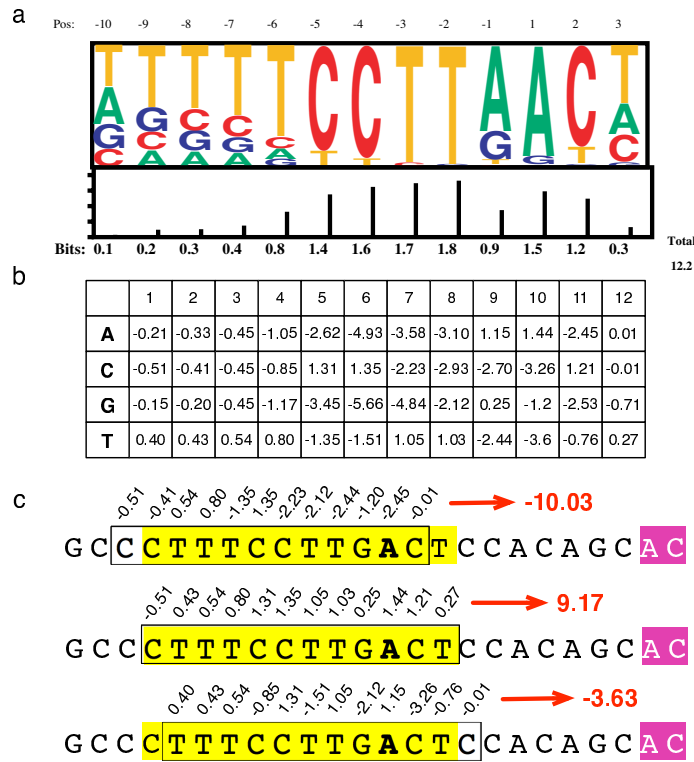


Fig. 2 Searching for signals. A position weight matrix (PWM) was calculated from known U12 branch point sequences. **(a)** The sequence logo shows the information content of the U12 branch point for human U12-dependent introns. **(b)** The PWM contains the log likelihood ratios (signal/background) for each base at each position of the 12bp profile. **(c)** A 12bp window is advanced one base pair at a time over the genomic sequence and the log ratios are summed over each position to give the branch point score. The result of scoring the positions immediately before, exactly over and immediately after the branch point are shown. The branch adenosine is shown in bold and the profile-matching bases are highlighted in yellow

Donor splice sites, for example, are often modeled as 1st or 2nd order Markov chains. In fact, so are acceptor splice sites, branch points, polypyrimidine tracts, and start sites, among other signals.

Sometimes, however, non-adjacent positions exhibit dependencies, for example in the donor site motif. Several methods have been developed to capture these dependencies. Maximal dependence decomposition (MDD), which is used by Genscan (Burge and Karlin, 1997), uses a decision tree to select one of several WAMs for scoring the site. Inclusion-driven learned Bayesian Networks (idIBNs) have also been tried (Castelo and Guigó, 2004). These methods outperform PWMs and first-order Markov models when predicting individual sites, but the improvements tend to vanish when considered in the overall framework of a gene finding program. Sup-

port vector machines (SVMs) trained with sequence features local to the splice site have also shown promise (Sun *et al.*, 2003; Zhang *et al.*, 2003; Degroeve *et al.*, 2005; Baten *et al.*, 2006; Rätsch *et al.*, 2006), however, it is unclear to what extent other features such as codon usage (usually detected separately from the splice site) influence their success. When used alone (not in a gene prediction context), they perform substantially better than the PWM or first-order Markov model (WAM).

2.2.2 Content

In theory, the signals on their own should completely specify the intron-exon structure of a transcript. However, proper classification of all potential start codons and splice sites in a genomic sequence is still a challenge. Properly detecting the start and end of transcription is also a major challenge. This suggests that either our models of these signals are inadequate, or we have yet to identify additional signals involved (such as cis-acting enhancer or silencer elements affecting splice site choice), or our models of the mechanisms of transcription and/or splicing are deficient or a combination of all of the above. Therefore, most gene prediction strategies also take advantage of the statistical properties of coding sequences. We call such content-based coding versus non-coding measures “coding statistics.”

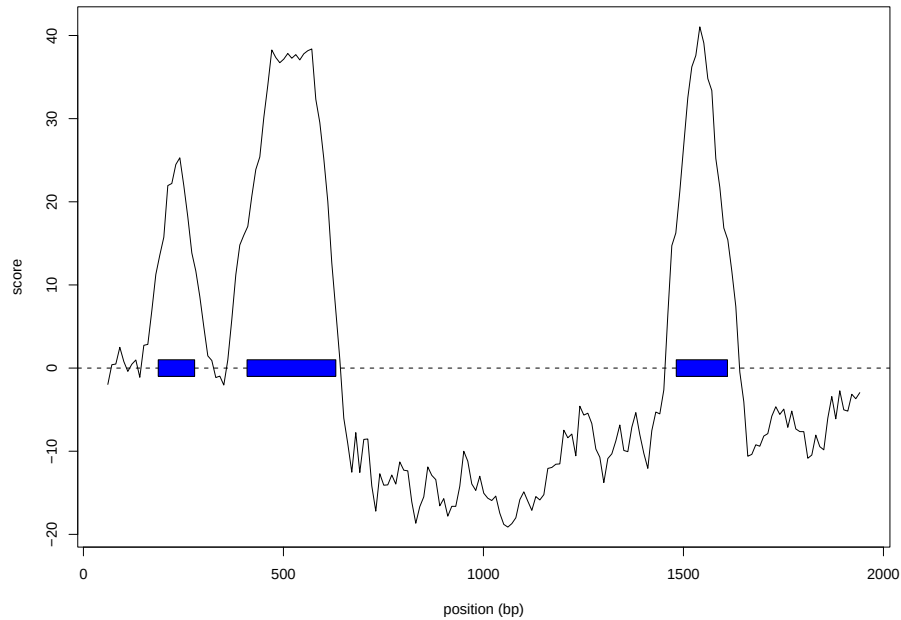


Fig. 3 Coding potential calculated using a fifth-order Markov model over the human beta globin gene locus. Annotated exons are shown in blue

Indeed, protein coding regions exhibit characteristic DNA sequence composition bias, which is absent from non-coding regions (see Figure 3). The bias is a consequence of (1) the uneven usage of the amino acids in real proteins, and (2) of the uneven usage of synonymous codons. To discriminate protein coding from non-coding regions, a number of content measures can be computed to detect this bias (Fickett and Tung, 1992; Gelfand, 1995; Guigó *et al.*, 2000). Such coding statistics can be defined as functions that compute a real number related to the likelihood that a given DNA sequence codes for a protein (or a fragment of a protein). Most coding statistics measure directly or indirectly either codon or di-codon usage bias, base compositional bias between codon positions, or periodicity in base occurrence (or a mixture of them all). Since the early eighties, a great number of coding statistics have been published in the literature. Hexamer frequencies usually in the form of codon position dependent 5th-order Markov models (Borodovsky and McIninch, 1993) appear to offer the maximum discriminative power, and are at the core of most popular gene finders today. In practice it is implemented as a three-periodic inhomogeneous Markov model, with one Markov chain corresponding to each position of a codon. GRAIL (Uberbacher and Mural, 1991; Xu *et al.*, 1994), an earlier gene finding method, popular in the early nineties, used a neural networks to determine the optimal combination of a variety of coding statistics for predicting coding regions.

2.3 Conservation

When one or more informant genomes are available, it is possible to detect the characteristic conservation pattern of coding sequence and use it as an orthogonal measure of coding potential. Over the past few years, several programs have been developed that exploit sequence conservation between two genomes to predict genes. A wide variety of strategies have been explored. In one such strategy (Alexandersson *et al.*, 2003) (further discussed below), alignment of the genomic sequence and gene prediction are performed simultaneously. In the “informant genome” approach (e.g. SGP2 (Parra *et al.*, 2003) and TWINSKAN (Korf *et al.*, 2001) alignments are performed first using standard tools such as TBLASTX or BLASTN and these alignments are used to inform prediction. More recently methods that use multiple alignments among several genomes have been developed.

To illustrate this point, in Figure 4 we display the human beta globin gene locus on the UCSC genome browser. The definitive annotation is represented by the aligned RefSeq sequence at the top, while the conservation track at the bottom shows the evolutionary conservation as determined by a phylo-HMM. In between are various gene predictions which use 0 (GeneID), 1 (SGP2) or 27 (CONTRAST) aligned genomes.

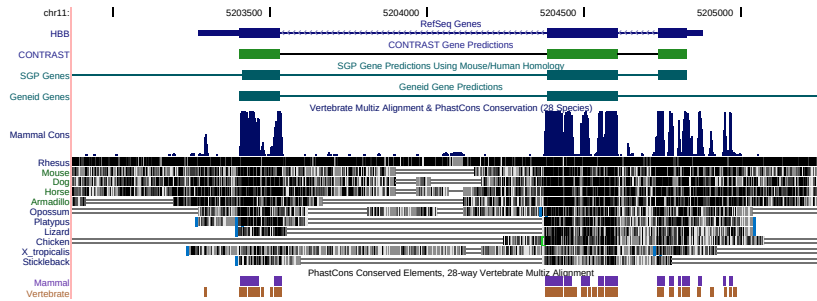


Fig. 4 Coding sequences are more conserved than non-coding sequences. Conservation within mammals at the human beta globin gene locus is shown. Gene prediction programs that utilize conservation (CONTRAST and SGP2) perform better than those that do not (GeneID)

3 Frameworks for Integration of Information

As we have seen, genomic and extra-genomic information of many different forms (sequence motifs, coding nucleotide composition, evolutionary conservation) can contribute to the prediction of the intron-exon structure of protein-coding transcripts. Successful gene prediction, however, depends on more than the sum of its parts; accurate and efficient integration of this information is critical. In this section we will look at gene prediction from the perspective of integration, outlining the various frameworks that have been developed and elaborated over the years.

3.1 Exon-chaining

Once exons are predicted, explicitly or implicitly, along a genomic sequence, exons need to be chained into gene predictions. Exon-chaining, therefore, is actually something that every gene predictor does, at least conceptually. The main difficulty in exon assembly is the combinatorial explosion problem: the number of ways N candidate exons may be combined grows exponentially with N . The key idea of computational feasibility comes from dynamic programming (DP), which allows finding the “optimal assembly” quickly without having to enumerate all possibilities (Gelfand and Roytberg, 1993). Exon chaining DP (Guigó, 1998) is implicit to several currently available gene predictors such as Fgenesh (Solovyev *et al.*, 1995) and GeneID (Guigó *et al.*, 1992; Parra *et al.*, 2000). In GeneID, gene prediction is done hierarchically. First, splice sites, start and stop codons are predicted and scored on the query sequence. From these sites, all potential protein coding exons are built. The exons are scored as a function of the scores of the exon defining sites, and the score of a fifth-order Markov model which evaluates the coding bias of the predicted exon sequence. Because in GeneID all scores are log-likelihood ratios, the score of the exons is simply the sum of individual scores. Finally, exons are assembled into

gene structures, so that the final assembly is the one maximizing the sum of the assembled exons.

The advantages of the hierarchical approach is that the gene finding problem can be tackled in discrete steps and analyzed at intermediate stages. It is also very fast and can analyze large mammalian genomes in only a few hours. It also allows for a quite flexible scoring approach, since exons can be re-scored, using ad-hoc procedures, depending on their conservation in other genome(s) or their similarity to known protein or cDNA sequences. However, a number of shortcomings are apparent, especially when compared to the more recent crop of HMM and CRF-based gene predictors (see below): exon and intron length distributions are not very well modeled (only minimum and maximum lengths can be specified), and scores are not truly probabilistic.

3.2 *Generative Models: Hidden Markov Models*

A novel advance in eukaryotic gene prediction methodologies was the application of generalized Hidden Markov Models (HMMs), initially implemented in the Genie algorithm (Kulp *et al.*, 1996) (HMMs were first used in a bacterium gene finder by Krogh, *et al.* (Krogh *et al.*, 1994) after its success in protein modeling.) Soon after, it was implemented in the Genscan algorithm (Burge and Karlin, 1997) to predict multiple genes. Several other HMM-based gene prediction programs were developed later: Veil (Henderson *et al.*, 1997), HMMgene (Krogh, 1997) and Fgenesh (Salamov and Solovyev, 2000).

In the HMM approach, different types of structure components (such as exons or introns) are characterized by a state, and the gene model is thought to be generated by a state machine: starting from 5' to 3', each base-pair is generated by an "emission probability" conditioned on the current state (and if using a higher order Markov model, a limited number of preceding bases), and the transition from one state to another is governed by a "transition probability" which obeys a number of constraints (e.g an intron can only follow an exon, reading frames of two adjacent exons must be compatible, etc.). All of the parameters of the emission probabilities and the (Markov) transition probabilities are learned (pre-computed) from some training data. Since the states are unknown ("hidden"), an efficient algorithm (called the Viterbi algorithm, similar to DP) may be used to select the best set of consecutive states (called a "parse"), which has the highest overall probability of any possible parse for the given genomic sequence without actually having to enumerate all possible parses (see (Rabiner, 1989) for a tutorial on HMMs).

The reason these fully probabilistic state models have become preferable is that all scores are probabilities themselves and the weighting problem becomes only a matter of counting relative observed state frequencies. It is easy to introduce more states (such as intergenic regions, promoters, UTRs, etc.) and transitions into HMM-based models to accommodate partial genes, intronless genes, even multiple genes

or genes on different strands. These features are essential when annotating genomes or large contigs in an automated fashion.

In the following sections, we will outline the various “flavors” of HMMs that have been applied to the problem of gene prediction, starting with the basic HMM.

3.2.1 Basic Hidden Markov Models

The first HMM-based gene predictors such as Genie were designed around a basic hidden Markov model, which is described by a set of possible states (e.g. start, exon, donor, intron, acceptor, stop, intergenic, etc.), a set of possible observations (e.g. the set of nucleotides A, C, G and T), a transition probability matrix, an emission probability matrix, and the initial state probabilities. Transition probabilities govern the chance of moving from one state to any of the other states (or even back to the same state), for example from an exon to a donor site, from a donor site to an intron, etc. Emission probabilities correspond to the frequencies of nucleotides occurring in particular states (similar to a PWM model).

For an example of a simple hidden Markov model that illustrates the concept of states, transition probabilities and emission probabilities, please refer to Figure 5, in which we show how one might design an HMM for detecting regions of high GC content. With this model, one can solve the following problems associated with an HMM:

1. *Evaluation.* Find the probability of the sequence given the model and its parameters. This would be the sum of all possible state paths through the sequence. The probability of one such path is shown in Figure 5b. To enumerate all possible paths and sum their probabilities is generally an intractable problem, however fortunately there exists a dynamic programming algorithm, the “forward” algorithm, that can solve it efficiently.
2. *Decoding.* Find the most likely state path (i.e. sequence of AT-rich and GC-rich regions) given the model and a particular sequence. This is solved by the Viterbi algorithm.
3. *Learning.* Adjust the parameters (initial, transition and emission probabilities) to maximize the likelihood of the sequence given the model. In the example in Figure 5, this would correspond to learning the probabilities of emitting the nucleotides A, C, G and T in each of the two states, AT-rich and GC-rich, and learning the probabilities of switching between the two states given a set of training sequences. If, however, the training sequences are already annotated with AT-rich regions, the learning step can be bypassed and the transition and emission probabilities set to the frequencies and base composition corresponding to the annotation.

Hidden Markov models for gene prediction, on the other hand, are necessarily more complex than the example in Figure 5 due to the larger number of states and possible transitions needed to model gene structures. The first step in gene finding using an HMM is to learn the parameters from either labeled data (i.e. known

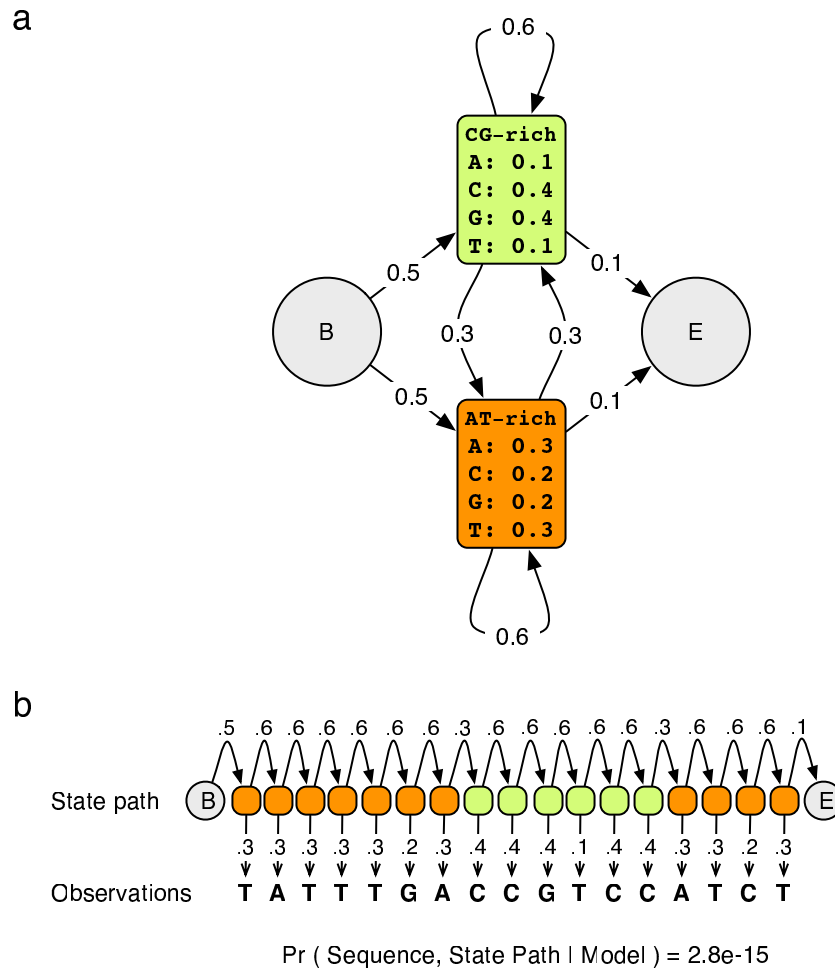


Fig. 5 A simple HMM for detecting regions of high GC content. **(a)** The state diagram exhibits two states which “emit” sequence according to different nucleotide probabilities. The begin (“B”) and end (“E”) states are silent, i.e. they do not emit sequence. Transition probabilities are shown with arrows. Transitions from one state to all others always sum to one. Emission probabilities are shown as tables in each of the two states. **(b)** The calculation of the joint probability $\Pr(x, y)$ of a sequence x and a particular state path or “parse” y is shown, and is simply the product of the transition and emission probabilities that were visited while traversing the path. The “true” sequence of states is “hidden”

genes) or unlabeled data. If the annotation is trusted, the transition and emission probabilities can simply be set to the frequencies observed in the annotated genes. Likewise, the weight array matrices for the various signals and content sensor sub-models that we described above are simply set by obtaining count frequencies. This procedure is called maximum likelihood estimation. In some cases, however, the optimal states are unknown, for example the ancestral evolutionary states in a phylo-HMM (described below). In these cases, the probabilistic basis of HMMs allows the parameters to be systematically learned from the data by maximum likelihood using the Baum-Welch algorithm (Baum, Leonard E. *et al.*, 1970), which is a special case of the Expectation Maximization algorithm (Dempster, A. P. *et al.*, 1977).

Once the model is trained, the software can be run on genomic sequences. Given the DNA sequence and the HMM model, a dynamic programming algorithm called the Viterbi algorithm can be used to find the optimal parse (i.e. the most likely sequence of exons and introns), or in other words annotate the sequence.

For gene finding, the probability of a sequence given an HMM is rarely solved for explicitly, although once an optimal path (in this case, a sequence of exons and introns) is predicted, its probability can give tell us something about how well it fits the model. The “Forward” and “Backward” algorithms are used to make this calculation.

3.2.2 Generalized Hidden Markov Models

One problem with the basic HMM is that the duration of a state can only be modeled as a transition back to itself with transition probability p . This in effect limits the duration of state to a geometric length distribution $E[l_X] = 1/(1 - p)$

In a generalized HMM (GHMM), length distributions can be explicitly modeled, for example with a Poisson point process, which is a counting process that represents the total number of occurrences of discrete events during a temporal/spatial interval. An additional variable d is introduced into the HMM. Upon entering a state, a duration is chosen according to a particular probability distribution and then d number of characters are emitted according to the emission probabilities. The transition to the next state is made according to the transition probabilities. The advantage of this is that exon lengths and intron lengths can be explicitly modeled according to their estimated length distributions obtained from training. The disadvantage is an increase in computational complexity, thus often compromises are made. The program Augustus (Stanke *et al.*, 2006), for example, reduces this computational cost by explicitly modeling short introns and using a geometric distribution for longer introns.

Another advantage of GHMMs is that they are modular. The states, in fact, can be represented by any suitable model and can be trained separately from the main model. For example, in Genscan, one of the first programs to utilize a GHMM, the donor site is modeled using maximal dependence decomposition (MDD) while the acceptor site is modeled by a standard Markov chain. Such modularity facilitates the design of the overall gene model, allowing one to easily incorporate additional

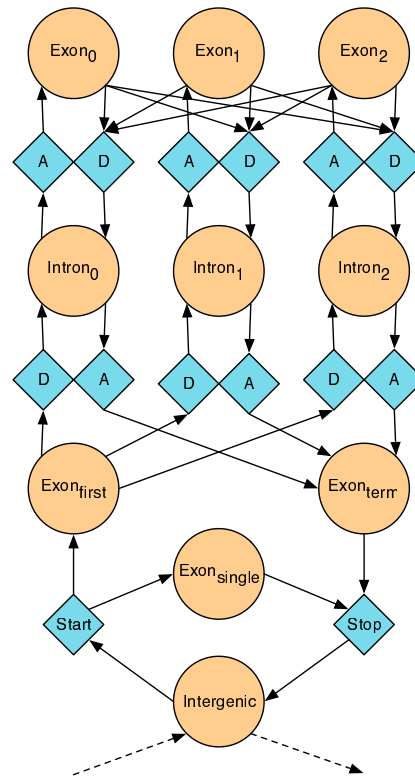


Fig. 6 A typical state diagram for a generalized hidden Markov model used for eukaryotic gene-finding. The three intron phases/exon frames are modeled by the separate intron and exon states 0, 1 and 2. Signal states donor (D), acceptor (A), start codon and stop codon (diamonds) mark the transitions between the variable-length content states introns, exons and intergenic regions (circles). Only the states for plus strand prediction are shown; simultaneous minus strand prediction are handled by a mirror image of the states linked through the intergenic state (not shown)

states. A basic state diagram for gene prediction is shown in Figure 6. There are usually separate models for each intron phase and exon frame, thus enabling proper frame consistency.

3.2.3 Generalized Pair HMMs

As described above in Section 2.3, the availability of multiple fully sequenced genomes heralded the advent of multi-genome *de novo* gene predictors. SGP2 directly uses BLAST scores to modify the log odds that a particular candidate exon is coding. Twinscan modified the Genscan model to use an extended alphabet (8 characters) corresponding to aligned and unaligned versions of the four bases, A,

C, G and T. This represented a precursor to the next class of HMMs called generalized pair HMMs, pioneered by the program SLAM (Alexandersson *et al.*, 2003) and also utilized by the program TWAIN (Majoros *et al.*, 2005). Generalized pair HMMs (GPHMMs) represent a fully probabilistic comparative genomic approach that simultaneously produces both an alignment and annotation of two syntenic regions. Pair HMMs have traditionally been used in pairwise alignment algorithms and include match, insert and gap states. A generalized pair HMM is similar in that it emits gene features as aligned pairs (exon pairs or intron pairs, for example, one in each species). In addition to the set of parameters required by GHMMs, the GPHMM is additionally specified by a joint distribution of paired durations and a joint distribution of pair emission probabilities. A parse then becomes a series of states with paired durations. In general, exon insertion/deletions are not allowed, although Doublescan (Meyer and Durbin, 2002), which uses a non-generalized pair HMM, does allow for indels.

The advantages of using GPHMMs are first of all, increased accuracy compared with methods that utilize only a single genome, and second you get two predictions for the price of one – gene predictions are made simultaneously in both genomic sequences. However variability in exon number is not tolerated, there are more parameters to estimate and the requirement for lengthy stretches of syntenic sequence is often difficult to meet, making their use in practice somewhat limited.

3.2.4 Phylo-HMMs or Evolutionary HMMs

If a whole genome alignment of more than one genome is available, it is possible to integrate this information into a gene-finding HMM by explicitly modeling the evolutionary history of the DNA sequence. Phylo-HMMs (Siepel and Haussler, 2004) (also called evolutionary HMMs (Pedersen and Hein, 2003)) model a combination of two Markov processes operating in two different dimensions: space (along a genome, like in traditional GHMM gene finding) and time (along the branches of a phylogenetic tree.) Basically, the columns of a multiple alignment are emitted according to a complex phylogenetic model such as the nucleotide substitution model of Hasegawa, Kishino and Yano (HKY) (Hasegawa *et al.*, 1985), which is modeled using a continuous time Markov chain. The probability of mutation at a particular site is allowed to depend on the pattern of mutation at the previous few sites (obeying the Markov assumption) and the evolutionary rate in general differs according to biological function (coding versus non-coding, for example) and can also be allowed to vary from one region of the genome to another.

The UCSC conservation track is probably the best known example of a phylo-HMM. This model has also been successfully implemented in the gene prediction programs Shadower (McAuliffe *et al.*, 2004) and N-SCAN (Gross and Brent, 2006), a multi-genome version of Twinscan.

Phylo-HMMs represent a true advancement in the integration of multi-genome conservation and performance gains are seen over single- and dual-genome predic-

tors. However, their use is restricted to cases where well-aligned genome sequences exist, and their computational cost is quite high.

3.3 Discriminative Learning

Hidden Markov model based gene prediction has represented the state of the art of eukaryotic gene prediction for many years. More recently, however, we are beginning to see the application of new theoretical frameworks which may be best classified as discriminative in nature, as opposed to the generative nature of HMMs. In discriminative learning, the posterior probability $Pr(y|x)$ of hidden states (gene structure) given the observations (DNA sequence) is modeled directly. In generative learning (HMMs), a more general problem, estimation of the joint probability $Pr(x,y)$ of the states and observations from training data (as in Figure 5b), is solved before calculating the posterior probability $Pr(y|x)$ according to Bayes rule (Ng and Jordan, 2001), where x corresponds to the observations and y corresponds to the labels or state path.

The direct modelling of the probability of a gene annotation (a sequence of labeled segments, i.e. state path) given a sequence (the observations) lends itself to discriminative training, a training paradigm in which all parameters of the model are tuned or weighted in order to directly maximize the discriminatory power of the model. In the case of gene prediction, this means determining the weights of various model parameters in order to achieve maximum annotation accuracy according to defined measures of gene prediction accuracy (see Section 5). This type of training, in which the model is trained to maximize a conditional probability $Pr(x|y)$ versus a joint probability $Pr(x,y)$, is also called “conditional training.” Semi-Markov (or generalized) versions of support vector machines (SVMs) and conditional random fields (CRFs), both discriminative in nature, are promising newcomers to the field of gene prediction.

3.3.1 Support Vector Machines

Support vector machines (SVMs), a particular set of supervised learning methods, have rapidly become popular in biological research to solve classification problems. SVMs are designed to discriminate two classes, for example true splice sites from decoy sites, by separating them with a large margin. SVMs are trained by learning this margin, or boundary, from positively and negatively labeled training examples.

SVMs for gene prediction have been independently applied to the problems of splice site detection and exon content (coding versus non-coding) classification; however, more recently, the SVM framework has been generalized and applied to the exon assembly problem, resulting in the programs mSplicer and mGene (Rätsch *et al.*, 2007). Briefly the scores of the signal and content submodels (themselves learned by SVMs) are combined with segment length contributions and then given

to piecewise linear weighting functions which have been trained to maximize the margin between the score of the best gene model and that of all false models.

3.3.2 Semi-Markov Conditional Random Fields

Most recent on the scene of eukaryotic gene prediction are a set of programs based on semi-Markov conditional random fields (SM-CRFs). A SM-CRF on a sequence x outputs a segmentation of x in which labels are assigned to segments of the sequence (e.g. exon, intron, etc.) They are essentially “conditionally trained” semi-Markov chains, that is, they are designed to find the most likely set of labels (states) that the model has been trained to traverse given a set of observations (input sequence). SM-CRFs are analogous to GHMMs (or semi-HMMs) except that the probability of label-value pairs, the labels being conditioned on the values, is learned directly. The values or observations are examined and not “emitted” as they are in HMMs, which in many respects is more intuitive and more accurately reflects the problem that is trying to be solved. Some advantages of this framework are that (1) any discriminative feature corresponding to an arbitrary-length segment may be used, (2) it need not be probabilistic and (3) features may overlap – discriminative training will assign appropriate weights.

Recent examples of semi-Markov CRF implementations for gene prediction include:

- CRAIG (Bernal *et al.*, 2007), which is trained globally on all input feature vectors using an online large-margin algorithm related to multiclass SVMs
- CONRAD (DeCaprio *et al.*, 2007), which is provided as a generic gene calling engine that promises to be highly customizable, although it has only been trained so far on fungal species
- CONTRAST (Gross *et al.*, 2007), a multi-genome predictor that is “phylogeny free” working directly with features extracting from whole genome multiple alignments.

The semi-Markov CRF framework would appear to hold much promise for the integration of multiple sources of information and may become the *de facto* model for such purpose.

3.4 Combiners

Programs that specifically aim to integrate the results of other gene callers have been dubbed “combiners.” Previous work has produced many such programs: GAZE (Howe *et al.*, 2002), Jigsaw (Allen and Salzberg, 2005), GLEAN (Elsik *et al.*, 2007), Genomix (Coghlan and Durbin, 2007), and EuGène (Foissac and Schiex, 2005) to name a few. The goal of such programs is to automate the task that faces human an-

notators: to produce an annotation when presented with the results of many different and potentially conflicting gene predictions.

While the combining functions differ among programs, the general principle on which they operate is that predictions should make uncorrelated errors which should tend to cancel each other out and increase the signal to noise ratio. This principle relies on the assumption that the input predictions are independent. However, this is often not the case due the use of similar methods, training data or extrinsic evidence. This can be circumvented by careful choice of input methods or can be explicitly corrected for by the combining algorithm, as is done by the combiner GenePC, which we are developing.

In general, combiners perform better than any individual input, often dramatically improving on specificity measures at all levels. For this reason, they are becoming popular for the automated annotation of new genomes.

4 Training

In most gene prediction programs, there is a clear separation between the gene model itself and the parameters of the model. While the model is general, the parameters often need to be specifically estimated for different species, or taxonomic groups. Using the wrong parameters may lead to mispredictions. Typically, the parameters of the gene model define the characteristic of the sequence signals involved in gene specification (i.e. weight matrices for splice sites), the codon bias characteristic of coding exons (i.e. hexamer counts or Markov models for coding regions), and the relation between the exons when assembled into gene models (i.e. intron and exon lengths distributions, number of exons, etc.). These parameters are estimated from a set of annotated genomic sequences from the species of interest. If not enough annotated sequences are available, some programs, such as GLIMMER (Salzberg *et al.*, 1998), allow for the use of Markov models of smaller order.

Depending on the framework, the exact training algorithms differ from program to program. However, almost all gene predictors end up being trained discriminatively as some point to fine tune the model parameters (both submodel and global parameters) in order to achieve maximum discrimination – and it seems that all programs are characterized by the presence of “fudge” factors that get manually tuned regardless of the training procedure used. For example, we have mentioned above that HMM-based predictors can be trained using the Baum-Welch EM algorithm; however, such maximum likelihood training is usually performed on each submodel separately and then the global model tuned afterwards, usually manually. It has been shown that further improvements are realized when formal discriminative training methods such as generalized gradient ascent are used so as to maximize mutual information (MMI) on all the model parameters at once (Majoros and Salzberg, 2004).

Because of all these and other reasons, training a gene prediction program for a new species or taxonomic group is not always a trivial exercise; it requires a lot of manual intervention, and very few applications, if any, offer automatic training

protocols. Recently, however, methods have been developed to train gene-finding software even in the total absence of annotated genomic sequences of the organism under consideration (an increasingly common problem, when the sequencing of the genome of an organism is not followed by the sequencing of cDNAs from that organism) (Lomsadze *et al.*, 2005; Korf, 2004).

A limiting amount of training sequence available can also impinge on evaluation procedures (described in the next section). Of course it is desirable to train on as many known genes as possible to avoid overfitting; however, evaluation of the performance of a program should always be carried out on a clean set of genes on which the program's parameters were not estimated, in order not to bias the results. This is especially true when the model is trained to achieve maximum discriminative power. In this case one can perform an N-fold cross-validation or jackknife procedure, in which successive rounds of training and evaluation are performed with some of the data for training withheld and used for evaluation purposes. The results of all the rounds are then combined to give the final performance values.

5 Evaluation of Gene Prediction Methods

5.1 The Basic Tools

Whether running gene prediction pipelines, or just running gene prediction programs on a locus of interest, it is important to compare the outputs of multiple runs of a predictor with different settings or to compare multiple predictions from different programs. The comparison should be able to tell you something about the quality of each prediction by graphically reflecting the confidence in each exon, and should be of sufficient resolution to compare alternative splice sites. Several solutions to this problem have emerged.

The program GFF2PS (Abril and Guigó, 2000) is a highly customizable UNIX-based script for generating postscript figures from multiple prediction outputs or annotations in GFF format. GBROWSE is a database-driven application that performs a similar but web-based function. Perhaps the most easy to use online system, provided your genome is represented and you know the genomic coordinates of your annotations, is UCSC Genome Browser's custom track option. If you are an annotation group and provide annotation to the scientific community on a regular basis then the Distributed Annotation System (DAS) is the preferred approach. The most used DAS client for gene prediction annotations is ENSEMBL.

5.2 Systematic Evaluation

In addition of having some clue on the accuracy of the predictions on particular cases, one would like to have an overall measure of the accuracy of the “*ab initio*” gene prediction programs. The accuracy of gene prediction programs is usually measured in controlled data sets. To evaluate the accuracy of a gene prediction program on a test sequence, the gene structure predicted by the program is compared with the actual gene structure of the sequence. The accuracy can be evaluated at different levels of resolution. Typically, these are the nucleotide, exon, and gene levels. These three levels offer complementary views of the accuracy of the program. At each level, there are two basic measures: Sensitivity (Sn) and Specificity (Sp), which essentially measure prediction errors of the first and second kind. Briefly, Sensitivity is the proportion of real elements (coding nucleotides, exons or genes) that have been correctly predicted, while Specificity is the proportion of predicted elements that are correct. More specifically, if TP are the total number of coding elements correctly predicted, TN, the number of correctly predicted non-coding elements, FP the number of non-coding elements predicted coding, and FN the number of coding elements predicted non-coding, then, in the gene finding literature, Sensitivity is defined as $Sn = TP / (TP + FN)$ and Specificity as $Sp = TN / (TN + FP)$. Both, Sensitivity and Specificity, take values from 0 to 1, with perfect prediction when both measures are equal to 1. Neither Sn nor Sp alone constitute good measures of global accuracy, since high sensitivity can be reached with little specificity and vice versa. It is desirable to use a single scalar value to summarize both of them. In the gene finding literature, the preferred such measure on the nucleotide level is the Correlation Coefficient defined as

$$CC = \frac{(TP \times TN) - (FN \times FP)}{\sqrt{(TP + FN) \times (TN + FP) \times (TP + FP) \times (TN + FN)}}$$

CC ranges from -1 to 1, with 1 corresponding to a perfect prediction, and -1 to a prediction in which each coding nucleotide is predicted as non-coding and vice versa.

At the exon level, an exon is considered correctly predicted only if the predicted exon is identical to the true one, in particular both 5' and 3' exon boundaries have to be correct. A predicted exon is considered wrong (WE), if it has no overlap with any real exon, and a real exon is considered missed (ME) if it has no overlap with a predicted exon. A summary measure on the exon level is simply the average of sensitivity and specificity. At the gene level, a gene is correctly predicted if all of the coding exons are identified, every intron-exon boundary is correct, and all of the exons are included in the proper gene.

One of the first systematic evaluations of gene finders was produced by Burset and Guig (Burset and Guigó, 1996). These authors evaluated seven programs in a set of 570 vertebrate single gene genomic sequences. At that time, average exon prediction accuracy $((Sn + Sp)/2)$ ranged from 0.37 to 0.64. A few years latter, Rogic et al. (Rogic et al., 2001) updated the analysis; the average exon accuracy

of the tested programs increased to values between 0.43 to 0.76, illustrating the significant advances in computational gene finding that occurred during the nineties. (See Guigó and Wiehe (Guigó and Wiehe, 2003) for a review on the accuracy of gene prediction programs in the late nineties).

The evaluations by Burset and Guigó, Rogic et al. and others suffered, however, from the same limitation: gene finders were tested in controlled data sets made of short genomic sequences encoding a single gene with a simple gene structure. These data sets are not representative of the complete genome sequences being currently produced. To address this limitation, and in the context of large genome and annotation projects, more complex community evaluation experiments have been carried out to obtain a more realistic estimation of the actual accuracy of gene finding programs.

5.3 *The Community Experiments*

Community experiments – experiments on which many groups all over the world participate simultaneously – are becoming popular in Bioinformatics to comparatively benchmark the status of the prediction tools in a given area. One of the most well-known is CASP, which stands for Critical Assessment of Techniques for Protein Structure Prediction, and which takes place every two years since 1994. CASP provides the research community with an assessment of the state of the art in the field of protein structure prediction. Protein structures that are either expected to be solved shortly or that have been recently solved, but not yet discussed in public, are used as targets for the prediction. Predictions submitted by groups worldwide are then evaluated and compared.

5.3.1 GASP

GASP, the Genome Assessment Project, was inspired by CASP, and took place in 1999 in the context of the *Drosophila* Genome Project. In short, at GASP, a genomic region in *Drosophila melanogaster*, including auxiliary training data, was provided to the community and gene finding experts were invited to send the annotation files they had generated to the organizers before a fixed deadline. Then, a set of standards were developed to evaluate submissions against the later published annotations (Ashburner *et al.*, 1999), which had been withheld until after the submission stage. Next, the evaluation results were assessed by an independent advisory team and publicly presented at a workshop at the Intelligent Systems in Molecular Biology (ISMB) 1999 meeting. This community experiment was then published as a collection of methods and evaluation papers in *Genome Research* (Reese *et al.*, 2000).

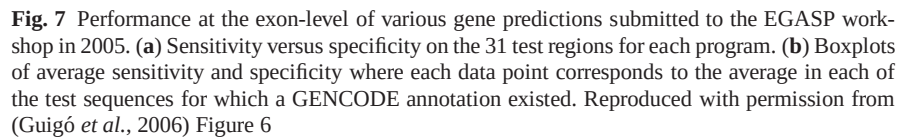
5.3.2 EGASP

Within the context of the pilot phase of the ENCODE project, the second GASP, the so-called ENCODE GASP (EGASP) took place. The 44 regions selected within the ENCODE project had been subjected to a detailed computational, experimental and manual inspection and a high quality gene annotation of the ENCODE regions had been produced – the so-called GENCODE annotation (Harrow *et al.*, 2006). On January 15, 2005 the complete gene map for 13 of the 44 regions was released, and gene prediction groups worldwide were asked to submit predictions for the remaining 31 regions. Eighteen groups participated, submitting 30 prediction sets by the April 15. The annotation of the entire set of the ENCODE regions was then released, and on May 6 and 7, participants, organizers and a committee of external assessors met at the Sanger Institute to compare the GENCODE gene map with the gene maps predicted by the participating groups. As with GASP, results were published as a collection of papers in the journal *Genome Biology* (Guigó *et al.*, 2006). Accuracy at the exon level for participating programs is shown in Figure 7. At EGASP some programs reached average exon accuracies close to 0.85.

5.3.3 NGASP

Very recently, NGASP, the nematode genome annotation assessment project, has taken place. Since five *Caenorhabditis* nematode genomes are currently available, those of *C. remanei*, *C. japonica* and *C. brenneri*, *C. elegans* and *C. briggsae*, nGASP was launched with the implicit goal of promoting the usage of the comparative information across these five genomes. The explicit goal was to objectively assess the accuracy of the current state of the art for protein-encoding gene prediction algorithms in *C. elegans*, and to apply this knowledge to the annotation of the other *Caenorhabditis* genomes. A set of regions representing $\approx 10\%$ (10 Mb) of the *C. elegans* genome was selected to evaluate the performance of the participating gene predictors. As with previous genome annotation assessment projects, participation was open to all academic, private sector, and government researchers. A summary of the results will be submitted for publication.

These community experiments are an excellent exercise to focus a whole community on a certain problem task and motivate groups and individuals to participate and submit their best possible solutions. External assessment of the results is critical and standards and rules have to be laid out clearly at the beginning of the experiment. They have been received with enthusiasm within the gene prediction community and they have had a great impact in tool development.



6 Discussion

6.1 Genome Datasets

There has been an effort to centralize all the information around the assembled sequences, and associated annotations, produced by the whole-genome sequencing projects. The best example are the three fully established whole-genome browsers: the NCBI Map Viewer (Wheeler *et al.*, 2001), the UCSC Genome Browser (Karolchik *et al.*, 2003) and the ENSEMBL browser at the Sanger Center (Hubbard *et al.*, 2002), each of which present by default a set of contributed gene-finding predictions from different programs obtained for each new released assemblies. In addition, each site develops its own “in-house” gene set. These sets are based on mRNA evidence obtained from cDNA and EST sequences, augmented with computational predictions.

ENSEMBL human genes are generated automatically by the ENSEMBL gene builder. They are of three basic types: those having full-length cDNA or proteins, those having high homology to proteins in other organisms and those Genscan-predicted genes matching to proteins/vertebrate mRNA and UniGene clusters. The basic gene-annotator engine (using protein homology to construct gene structure) is Genewise (Birney and Durbin, 2000). The “ENSEMBL genes” are regarded as being fairly conservative (with a low false positive rate), since they are all supported by experimental evidence of at least one form via sequence homology. Recently, ENSEMBL project has added spliced EST information for identification of alternative transcripts and to incorporate comparative genomics for getting orthologs and synteny relations. The basic annotator engine at the UCSC browser is BLAT (Kent, 2002) which allows rapid alignment of primate DNAs/RNAs or land vertebrate proteins onto the human genome reliably, hence annotating the genome by similarities. Finally, NCBI LocusLink has a rule-based genome annotation pipeline. Known genes are identified by aligning RefSeq genes (<http://www.ncbi.nlm.nih.gov/RefSeq/>) and GenBank mRNAs to the genome using MegaBLAST (Zhang *et al.*, 2000). Transcript models are reconstructed by attempting to settle disagreements between individual sequence alignments without using an a priori model (such as codon usage, initiation, or polyA signals). Genes (and corresponding transcript and protein features) are annotated on the contig if the defining transcript alignment is $\geq 95\%$ identity and the aligned region covers $\geq 50\%$ of the length, or at least 1000 bases. Finally, genes predicted by GenomeScan (Yeh *et al.*, 2001), an extension of Genscan to include protein homology information, are annotated only if they do not overlap any model based on a mRNA alignment.

6.2 Atypical Genes

Gene prediction efforts have been traditionally focused on predicting the “typical” gene. Genes with uncharacteristic features that do not appear with great frequency tend to be ignored such as, for example, genes possessing U12-type introns, selenoprotein genes with in-frame UGA codons which code for selenocysteine, fast-evolving genes or genes with atypical codon usage. Progress has been made in a couple of these cases.

U12 introns, which comprise only a fraction of a percent of all introns, are spliced by the minor spliceosome, a low-abundance spliceosome with a different composition of snRNPs than the major U2-dependent spliceosome. It binds to donor and branch point sequences which are highly conserved across all species in which they are found, which includes most animals, plants and even a few fungi and protists. However, the splice sites do not conform to the regular U2 consensus – they are quite divergent and many of them have AT-AC terminal dinucleotides, making them invisible to most gene prediction software. By incorporating WAMs for the U12 splicing signals into the GeneID parameter file and making a few modifications to the dynamic programming routine, we have made the latest version of GeneID able to predict genes with U12 splice sites without a significant decrease in specificity. To aid in future genome annotation efforts, introns from a wide range of eukaryotic genomes that have been classified as U12-type are now stored in a specialized database called U12DB (Alioto, 2007).

Selenoproteins pose an even greater challenge due to the presence of in-frame UGA codon(s) which are recognized by the selenocysteine tRNA in the presence of a SECIS element downstream, usually located in the 3' UTR. Yet these have also been systematically hunted down using a combination of *ab initio* gene prediction, RNA structure predictions and homology search (Kryukov *et al.*, 2003). The selenoproteome is now catalogued in the SelenoDB (Castellano *et al.*, 2008).

6.3 Outstanding Challenges to Gene Annotation

Community assessment experiments have revealed that computational methods are not able to reproduce the accuracy in the annotation that a dedicated team of annotators, evaluating the individual evidence that exist for the transcripts mapping to a given genomic locus, can produce. For instance, EGASP revealed that the most accurate of the gene finding programs are able to predict correctly only about 40% of the full length transcripts in the GENCODE annotation. The GENCODE annotation heavily relies on human supervision (by the HAVANA team at the Sanger Institute (Harrow *et al.*, 2006)) to solve the uncertainties arising from cDNA mapping onto the genome sequence, and it also includes computational predictions verified experimentally by RT-PCR and RACE. It is a much richer catalogue of the human transcriptome in the ENCODE regions than other existing gene sets. Indeed, the first release of the GENCODE annotation consisted of 2608 transcripts assigned

to 487 loci, more than doubling the number of alternative transcripts per locus in ENSEMBL. It looks like, therefore, there is still room for improving gene finding software that can automatically reproduce the task being carried out by human annotators when confronted with the complexity of transcription in the human genome.

This complexity, however, appears to be of a magnitude much higher than that implied by the GENCODE annotation. While extensive verification studies – including the EGASP community experiment (Guigó *et al.*, 2006) – have demonstrated that the GENCODE is essentially complete with respect to existing cDNA sequences and computational predictions, recent research by a number of groups using a variety of technologies shows that many transcripts exist that are not annotated in GENCODE. Indeed, data from high-throughput tag sequencing of cDNA ends (Shiraki *et al.*, 2003; Ng *et al.*, 2005; Peters *et al.*, 2007), from gene trapping in mouse embryonic stem cells (Roma *et al.*, 2007) and from hybridization of RNA samples into high density tiling arrays (Kapranov *et al.*, 2007; “The ENCODE Consortium”, 2007) reveals many additional sites of transcription. Particularly relevant are the results of the so-called RACEarray experiments in which the products of RACE reactions originating from primers anchored in exons from GENCODE genes are hybridized onto genome tiling arrays. More than half of the sites of transcription detected in this way (the so-called RACEfrags), which are by construction specifically linked to annotated protein coding genes, do not correspond to GENCODE annotated exons (Denoëud *et al.*, 2007). These results, therefore, are strongly indicative of the existence of a wealth of transcripts – including many alternative transcript forms of protein coding genes, and other transcriptionally complex events – which had so far escaped detection through systematic sequencing of cDNA libraries. Computational gene prediction methods are generally based on computational models that capture our understanding of the way proteins are encoded in genomes. Modeling these other types of transcripts may be far more challenging than modeling the standard protein-coding ones, as they may lack the strong signatures characterizing the latter.

6.4 What is the right gene prediction strategy?

The answer to the question of which gene prediction program to use is “all.” As of yet, no one program is even close to perfect, so the best advice is to run a handful of the best and combine their results using a gene prediction combiner. And even then, the gene models produced should be regarded as hypotheses about the gene structures embedded within the chromosome. These models can and should be validated by RT-PCR and/or direct sequencing.

While the state of the art in eukaryotic gene finding has improved steadily over the last decade, there is still a long way to go before we can automatically produce high-quality gene models for an entire genome, even one as well studied as the human genome. Moreover, the plethora of eukaryotic genomes being sequenced

now and in the future, and for which there is little transcriptional data, only increases the demand for better computational gene annotation methods.

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